Worksheet 7.1 — Equations of Lines

Grade 12 Calculus and Vectors (MCV4U) · Unit 7: Lines and Planes

The Big Ideas

A line needs a point and a direction vector $\vec{d}$. Vector: $\vec{r} = \vec{r}_0 + t\vec{d}$; parametric: $x = x_0 + ta, \ldots$; symmetric: $\frac{x-x_0}{a} = \frac{y-y_0}{b}$.

- Vector: $\vec{r} = \vec{r}_0 + t\vec{d}$.
- Parametric: $x = x_0 + ta,\ y = y_0 + tb\ (, z = z_0 + tc)$.
- Direction through two points: $\vec{d} = B - A$.

The Big Picture

Point and direction

A line is fixed by a point on it and a **direction vector**; the **vector equation** is $\vec{r} = \vec{r}_0 + t\,\vec{d}$.

Parametric form

Splitting into components gives the **parametric equations**, each coordinate written in terms of the parameter t.

2-D vs 3-D

In 2-D a line also has scalar and symmetric forms; in 3-D the vector and parametric forms are the natural description.

Worked Examples

Example 1 — Vector equation

Line through $(1, 2)$ with direction $(3, 4)$.

Solution. **Step 1 (recall the vector-equation form):** $\vec{r} = \vec{r}_0 + t\vec{d}$, using a point $\vec{r}_0$ and direction $\vec{d}$.

Step 2 (substitute the given point and direction): $\vec{r}_0 = (1, 2)$, $\vec{d} = (3, 4)$.

Conclusion: $\vec{r} = (1, 2) + t(3, 4)$.

Example 2 — Parametric

Parametric form of that line.

Solution. **Step 1 (split the vector equation into components):** $\vec{r} = (1, 2)+t(3, 4)$ means $x = 1+3t$ and $y = 2 + 4t$ separately.

Conclusion: $x = 1 + 3t,\ y = 2 + 4t$.

Example 3 — Through two points

Vector equation through $A(1,0,2), B(4,6,8)$.

Solution. **Step 1 (find a direction vector by subtracting the points):** $\vec{d} = B - A = (4-1,\ 6-0,\ 8-2) = (3,6,6)$.

Step 2 (pick either point, e.g. A, and write the equation): $\vec{r} = A + t\vec{d}$.

Conclusion: $\vec{r} = (1,0,2) + t(3,6,6)$.

Example 4 — Symmetric

Symmetric equation through $(1,2)$ with direction $(3,4)$.

Solution. **Step 1 (start from the parametric form):** $x = 1 + 3t \Rightarrow t = \dfrac{x-1}{3}$; $y = 2 + 4t \Rightarrow t = \dfrac{y-2}{4}$.

Step 2 (set the two expressions for t equal, eliminating the parameter): $\dfrac{x-1}{3} = \dfrac{y-2}{4}$.

Conclusion: $\dfrac{x-1}{3} = \dfrac{y-2}{4}$.

Example 5 — Point on the line?

Is $(4,6)$ on $\vec{r} = (1,2) + t(3,4)$?

Solution. **Step 1 (use the x-equation to solve for t):** $4 = 1 + 3t \Rightarrow t = 1$.

Step 2 (check the same t in the y-equation): $2 + 4(1) = 6$, which matches the given $y = 6$.

Conclusion: yes, $(4,6)$ is on the line (at $t = 1$).

Example 6 — Vector equation

Line through $(0,1)$ with direction $(2,5)$.

Solution. **Step 1 (apply $\vec{r} = \vec{r}_0 + t\vec{d}$):** $\vec{r}_0 = (0,1)$, $\vec{d} = (2,5)$.

Conclusion: $\vec{r} = (0,1) + t(2,5)$.

Example 7 — Parametric

Parametric form of $\vec{r} = (3,-1) + t(1,2)$.

Solution. **Step 1 (split into components):** $x = 3 + t$, $y = -1 + 2t$.

Conclusion: $x = 3 + t,\ \ y = -1 + 2t$.

Example 8 — Direction vector

Direction through $A(2,1,0), B(5,7,6)$.

Solution. **Step 1 (subtract A from B):** $(5-2,\ 7-1,\ 6-0)$.

Conclusion: $(3,6,6)$.

Example 9 — Point on the line?

Is $(7, 11)$ on $\vec{r} = (1, 2) + t(3, 3)$?

Solution. **Step 1 (solve for t using the x-equation):** $7 = 1 + 3t \Rightarrow t = 2$.

Step 2 (check the same t in the y-equation): $2 + 3(2) = 8$, but the given y-value is 11.

Conclusion: $8 \neq 11$, so no — $(7, 11)$ is not on the line.

Your Turn — Show Your Work

For each question, show all steps clearly in the space provided.

Question 1

Vector equation through $(0, 1)$ with direction $(2, 5)$.

Question 2

Parametric form of $\vec{r} = (3, -1) + t(1, 2)$.

Question 3

Direction vector through $A(2, 1, 0), B(5, 7, 6)$.

Question 4

Symmetric equation through $(0, 0)$ with direction $(1, 3)$.

Question 5

Is $(7, 11)$ on $\vec{r} = (1, 2) + t(3, 3)$?

Question 6

Vector equation through $(2, 3)$ with direction $(1, -1)$.

Question 7

Parametric form through $(1, 1, 1)$ with direction $(2, 0, 3)$.

Question 8

Direction vector through $A(0, 0, 0), B(3, 4, 12)$.

Question 9

Is $(5, 8)$ on $\vec{r} = (1, 2) + t(2, 3)$?

Question 10

Symmetric equation through $(2, 1)$ with direction $(4, 2)$.

Question 11

Vector equation of the x-axis.

Question 12

Is $(2, 4, 6)$ on $\vec{r} = (0, 0, 0) + t(1, 2, 3)$?

Question 13 — Challenge

Write the line through $A(1, 2, 3)$ and $B(4, 8, 9)$ in vector and parametric form.

Question Recap & Answer Key

Question Recap

1. Vector equation through $(0,1)$ with direction $(2,5)$.
2. Parametric form of $\vec{r} = (3,-1) + t(1,2)$.
3. Direction vector through $A(2,1,0), B(5,7,6)$.
4. Symmetric equation through $(0,0)$ with direction $(1,3)$.
5. Is $(7,11)$ on $\vec{r} = (1,2) + t(3,3)$?
6. Vector equation through $(2,3)$ with direction $(1,-1)$.
7. Parametric form through $(1,1,1)$ with direction $(2,0,3)$.
8. Direction vector through $A(0,0,0), B(3,4,12)$.
9. Is $(5,8)$ on $\vec{r} = (1,2) + t(2,3)$?
10. Symmetric equation through $(2,1)$ with direction $(4,2)$.
11. Vector equation of the x-axis.
12. Is $(2,4,6)$ on $\vec{r} = (0,0,0) + t(1,2,3)$?
13. Write the line through $A(1,2,3)$ and $B(4,8,9)$ in vector and parametric form.

✓ Answer Key

1. $\vec{r} = (0,1) + t(2,5)$
2. $x = 3 + t,\ y = -1 + 2t$
3. $(3,6,6)$
4. $\dfrac{x}{1} = \dfrac{y}{3}$
5. no
6. $\vec{r} = (2,3) + t(1,-1)$
7. $x = 1 + 2t,\ y = 1,\ z = 1 + 3t$
8. $(3,4,12)$
9. $5 = 1 + 2t \Rightarrow t = 2;\ 2 + 6 = 8 = 8$ yes
10. $\dfrac{x-2}{4} = \dfrac{y-1}{2}$
11. $\vec{r} = (0,0,0) + t(1,0,0)$
12. $t = 2$ throughout yes
13. $\vec{d} = (3,6,6);\ \vec{r} = (1,2,3) + t(3,6,6);\ x = 1 + 3t, y = 2 + 6t, z = 3 + 6t$